Data Preprocessing

Data pre-processing means transforming the raw data into a clean & structured format for prediction or analysis.

* Missing Value Detection & Treatment :-

Detection → Pandas → .isna() or .isnull()

Treatment → (1) Remove null values ⇝ only when there are few nulls

(2) fill null values ⇝ Mean, Median, Mode

Mean → Numerical Columns

Mode → Categorical Columns

Numerical Column → Mean → No Outliers

→ Median → Outliers

Categorical Column → Mode →

→ 0 or unknown

Name	Qual	Salary
A	MBA	●
B	–	●
C	MCA	●
D	MBA	●

* Outlier Detection & Treatment :-

Outlier → Extreme values in the dataset

Detection → (1) Boxplot

(2) Density Plot

(3) IQR

Treatment → (1) Removing Outliers ⇝ only when there are few outliers

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(2) Z-Score
(3) IQR
} 0, mean, median

★ Relationship b/w Multiple Variables :-

Numerical - Numerical ⇝ Correlation & Heatmap

Categorical - Categorical ⇝ Statistical Tests (Chi-squared Test)

Numerical - Categorical ⇝ " " (ANOVA)

★ Encoding the Categorical Columns :-

Encoding → Converting a categorical data into numerical data.

Methods for Encoding :-

1. One-Hot Encoding
2. Ordinal Encoding
3. Target Encoding
4. Custom Encoding

Nominal Encoding ⇝ When the data is unordered.

I. One-Hot Encoding :-

Quali.
B.Tech.
MBA
MCA

⇝

Qualif.	B.Tech.	MBA	MCA
B.Tech.	1	0	0
MBA	0	1	0
MCA	0	0	1

After encoding we drop the



After encoding we drop the qualification column

↓

Looked for unique values ⇝ Total unique values = 3 [BTech, MBA, MCA]

Drawback of OHE :-

- Increases the dimensionality of the data.

↓

no. of dimensions / no. of columns

The more the no. of dimensions we have, the difficult it is to train or handle the data.

↳ Complex data

↳ Time Consuming

↳ Memory Efficient

∴ When there are too many unique values in one column, we do not go for OHE.

II. Ordinal Encoding :-

It is used when we have ordered data.

Ex- Qualification, Rank, etc.

III. Target Encoding :-

It is used to encode the categorical columns based on the

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City	Salary
Delhi	70000
Mumbai	60000
Pune	45000
Delhi	62000
Pune	71000
Bang.	79000

↓

Look for unique values

① Delhi ⇝ 70000, 62000
② Mumbai ⇝ 60000
③ Pune ⇝ 45000, 71000
④ Bangalore ⇝ 79000

Mean/Median of individual group value is calculated

① Delhi ⇝ 66000
② Mumbai ⇝ 60000
③ Pune ⇝ 58000
④ Bangalore ⇝ 79000

→

City	Salary
66000	70000
60000	60000
58000	45000
66000	62000
58000	71000
79000	79000

IV. Custom Encoding :-

Encoding as per our preference

* Feature Engineering :-

It involves :-

- Creating New features
- Drop unimportant features
- Transforming existing features

★ Scaling :-

It is used to bring down all the values in a specific range.

Two types of scaling :-

(1) Min-Max Scaling [Normalization] :-

- Formula :- $$X_{new} = \frac{X_{old} - X_{min}}{X_{max} - X_{min}}$$
- It is applied when there are no outliers
- Range ⇝ 0 to 1

(2) Standard Scaling [Standardization] :-

- Formula :-

$$X_{new} = \frac{X_{old} - \mu}{\sigma}$$

μ = mean of column
σ = std. of column

- It can be applied / used in case of outliers
- Range ⇝ -3 to +3

- All the above are the data preprocessing steps.
- After carrying out the above steps, the data is finally prepared for ML model.

★ Ques. When to split the data into training or test dataset ?

Ans ⇝ Before pre-processing → Before missing value / outlier treatment
↳ Before label encoding

Treatment

↳ Before target encoding

↳ Before scaling

↳ Before Feature Engg. (depends)

Ques. Why before pre-processing ?

Ans → To avoid target leakage / data leakage.

		Age
Train	1	27
	2	35
	3	30
	4	29
	5	NaN
Test	6	31
	7	25

29.5

Fill NaN with mean = 29.5

↳ Now after filling NaN, if we split the data

↙ 5 rows as training

↘ 2 rows as test

Here, we did target leakage because while Calculating the mean, the values from target set was involved.

But we want to keep every info. from the target set to be hidden.

That's why we split the data before pre-processing.